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DLM lab practical 7

Q.1)

* Aim:- To design full subtractor using 74153 MUX IC and verify truth table.

* Theory:-
Adders and subtractors are essential in modern IC's. especially in ALU's & DSP units.

1. Half subtractor - A half subtractor is a combinational circuit that subtracts two binary bits sharing.
Input - A, B
output - S, Borrow.

2. Full subtractor - It considers Bin.
Input - A, B, Bin
output - S, Borrow.

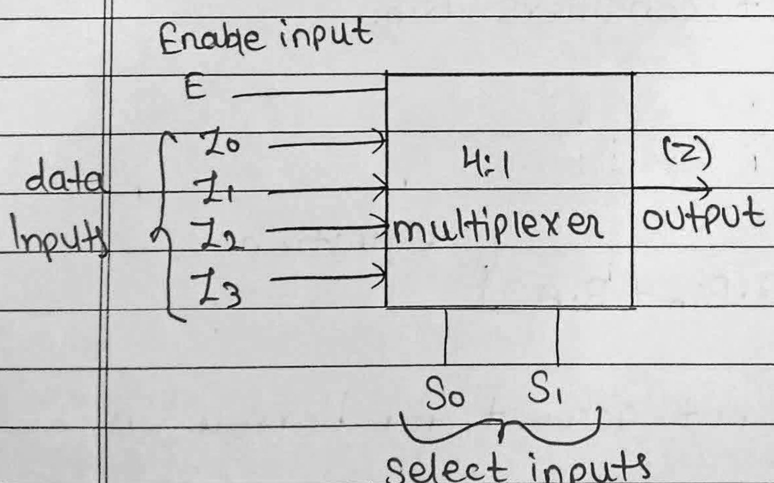
$$\left.\begin{aligned} S &= A \oplus B \oplus Bin \\ Borrow &= A \cdot Bin + B \cdot Bin + B \cdot A \end{aligned}\right\} \text{equations.}$$

→ Half subtractor does not account for borrow-in from previous stages.
→ But full subtractor consider borrow-in that's its one of the advantage over half.

Input			output	
A	B	Bin	S	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

* Multiplexer.

Combinational circuit which selects binary info present on any one of the input, depends upon logic status of the selection inputs, and routes it to the output line. If there are in selection lines, then the number n of max possible input lines is 2^n and the multiplexer is referred to as n to 1 multiplexer. A 4:1 multiplexer, has 4 input lines.



Select lines		output
S_0	S_1	Z
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

* Implementation of a Full Subtractor using 4:1 Multiplexer.

$$D(A,B,B_{in}) = \Sigma(1,2,4,7)$$
$$B_{out}(A,B,B_{in}) = \Sigma(1,2,3,7)$$

In view of these the output table of full subtractor can be written as.

	Input			output in terms of Bin	
	A	B	Bin	Difference (D)	Borrow Bout
1.	0	0	0	Bin	Bin
2.			1		
3.	0	1	0	Bin'	1
4.			1		
5.	1	0	0	Bin'	0
6.			1		
7.	1	1	0	Bin	Bin
8.			1		

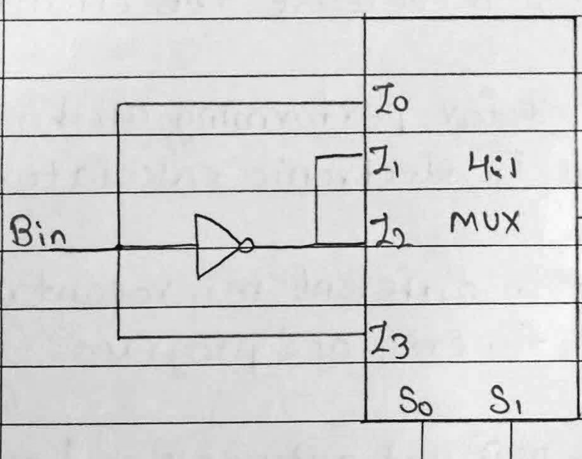


Fig: Difference output of a full subtractor with 4:1 mux

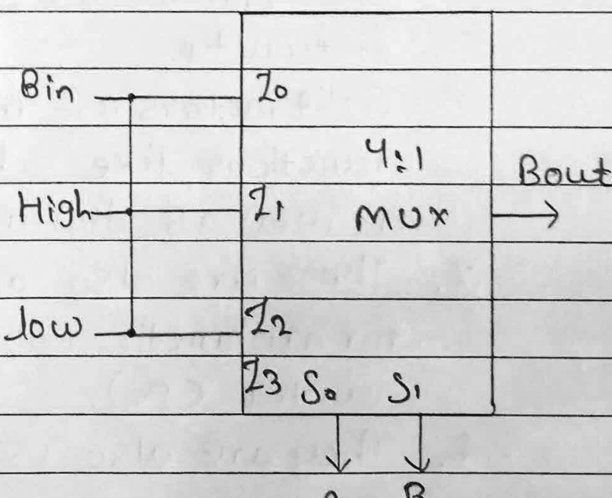


fig: Borrow output of a full subtractor with 4:1 MUX

* Function table of 74153 MUX IC for full subtractor.

Select I/P		Inputs (a)				Inputs (b)				output			
S_0	S_1	I_{0a}	I_{1a}	I_{2a}	I_{3a}	I_{0b}	I_{1b}	I_{2b}	I_{3b}	Z_a		Z_b	
A	B	Bin	Bin'	Bin'	Bin	Bin	1	0	Bin	D		Bout	
0	0	0	1	1	0	0	1	0	0	I_{0a}	0	I_{0b}	0
0	0	1	0	0	1	1	1	0	1		1		1
0	1	0	1	1	0	0	1	0	0	I_{1a}	1	I_{1b}	1
0	1	1	0	0	1	1	1	0	1		0		1
1	0	0	1	1	0	0	1	0	0	I_{2a}	1	I_{2b}	0
1	0	1	0	0	1	1	1	0	1		0		0
1	1	0	1	1	0	0	1	0	0	I_{3a}	1	I_{3b}	0
1	1	1	0	0	1	1	1	0	1		0		1

* Applications of Full Subtractor.

1. These are generally employed in ALU (Arithmetic Shift Unit) in computer to subtract and in CPU and GPU for the applications of graphics to decrease the circuit difficulty.
2. Subtractors are mostly used for performing arithmetical functions like subtraction, in electronic calculators as well as digital devices.
3. These are also applicable in different microcontrollers for arithmetic subtraction, timers, and program counter (PC).
4. They are also useful for DSP and networking based systems.

* Procedure.

Step:1) fill the truth table.
Step:2) Click on the component button to place the component.
Step:3) To design the full subtractor circuit, use the circuit diagram of the Ic or use connection table provided.
Step:4) connect outputs D and Bout to LEDS labelled D and Bout respectively.
Step:5) Once the connections are made, click on 'check connections' button. If connections are right, the 'Start Simulation' bottom will become active. Click on it to start the Simulation and use the input toggle switches marked 'A', 'B' and 'Bin' and observe the output.

* Conclusion:

The experiment successfully designed a full subtractor using the 74153 multiplexer Ic. The outputs matched the expected truth table, confirming the accurate implementation of the full subtractor and the effective use of the multiplexer in the logic design.